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Senior Capstone - Data Science
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Video Game: Project Egress

Project Proposal:

Our plan is to make a video game titled Project Egress. Project Egress will be a 2-D game that is similar to Portal where you pass through multiple short rooms or levels. The player character will have different ‘paints’ and weapons or tools to use said paints in order to reach the goal set in each level. Each weapon will provide a different method to splatter the paint and each paint will have a different effect.

More specifically, our game will be a puzzle-solving game, a platformer, a shooter, and a sidescrolling game. There should be roughly 3 to 4 different paint colors that will have different effects such as green being sticky, red being slippery or fast, and orange being bouncy. For the weapons, each weapon will fire a different size projectile and different rates (balancing) and we might even include entirely different tools all together if possible. We will be using ‘WASD’ for movement and left click for shooting. Our planned keybinds are entirely subject to change.

We will make our project with Unity which uses C#. In terms of roles, Ethan will be making the art and models for the game using a pixel art website which will later be ported to unity. Liam will be doing the majority of the coding for the game. Madeline will be helping with both the code and art. Our milestones will be: finishing a 2D sandbox environment, finishing pixel art, creating basic objectives, applying the code and sprites, completing the game.

Our main goal right now is to gather all of the resources we will need in order to learn exactly how to code and get the art to function properly. After that, our main goal will be to make the game itself, as listed in the previous paragraph. Overall, our main goal is to at least have a functioning game regardless of whether we polish it to the extent of having fully functional levels and a clean user interface. It is highly possible that we will achieve a sandbox, it is just a matter of how much time it will take to achieve everything. Our main priority is to be able to get all of our basic functions and art in the game so it is still enjoyable to play.

Languages/Software:

Languages:

- C#
- GDScript

Software (Good Options):

- Unity
- Godot

Links/Resources:

Coding:

- 2d unity: <https://learn.unity.com/tutorial/2d-game-kit-walkthrough>
- 2d sprite rigging:
<https://learn.unity.com/tutorial/rigging-a-sprite-with-the-2d-animation-package>
- 2d asset importing:
<https://learn.unity.com/tutorial/importing-2d-assets-into-unity-2019-3>
- Unity: <https://www.youtube.com/watch?v=XtQMytORBmM>
- C# intro: <https://www.codecademy.com/learn/learn-c-sharp-introduction>

Art:

- Pixel art maker: <https://www.piskelapp.com/>
- Animated Sprites: <https://www.youtube.com/watch?v=7AKGb2sg19E>

Taskbar icon CONCEPTS:

